Lesson 3.14 - Mean, Median, and Mode of Grouped Data

> **Video:** Mean, Median, and Mode of Grouped Data & Frequency Distribution Tables by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

Idea

With **grouped data** you only know how many values fall in each **class interval**, not the individual values. The mean can only be **estimated**. Each class is represented by its **midpoint**:

$$\text{midpoint } m = \frac{\text{lower limit} + \text{upper limit}}{2}$$

$$\bar{x} \approx \frac{\sum f\,m}{\sum f}$$

- **Modal class:** the interval with the highest frequency.
- **Median class:** the interval containing the middle value, at position $n/2$. Find it with a **cumulative frequency** column.

Example 1 - test grades

Grade	f	Midpoint m	$f \cdot m$	Cumulative f
40-49	3	44.5	133.5	3
50-59	5	54.5	272.5	8
60-69 (D)	6	64.5	387.0	14
70-79 (C)	9	74.5	670.5	23
80-89 (B)	8	84.5	676.0	31
90-100 (A)	7	95.0	665.0	38
Total	**38**		**2804.5**	

- Midpoint example: $(40 + 49)/2 = 44.5$. The last class is $(90 + 100)/2 = 95$.

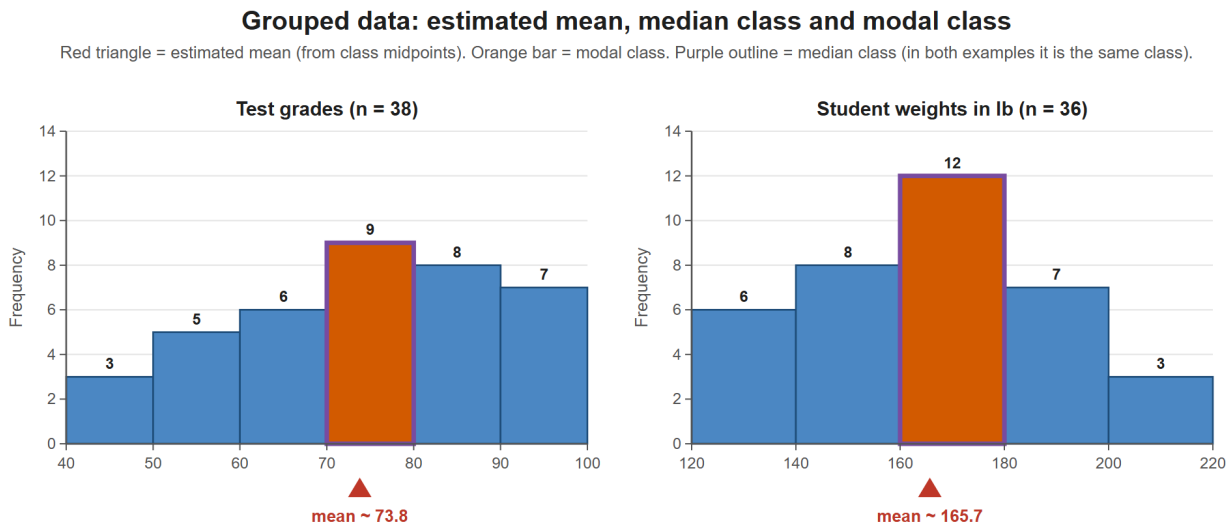
Mean:

$$\bar{x} \approx \frac{\sum fm}{\sum f} = \frac{2804.5}{38} \approx 73.8$$

This is reasonable: the most common class is 70-79, and the estimate lands there.

Mode: the highest frequency is 9, so the modal class is **70-79**.

Median: half of 38 is 19. The cumulative counts reach 14 by the end of 60-69 and 23 by the end of 70-79. The 70-79 class holds students 15-23, which includes 19, so the median class is **70-79**.



The estimated mean (red triangle) is computed from class midpoints. In both examples the class with the highest bar (the modal class) also contains the middle value (the median class).

Example 2 - student weights (lb)

Weight	f	Midpoint m	$f \cdot m$	Cumulative f
120-139	6	129.5	777.0	6
140-159	8	149.5	1196.0	14
160-179	12	169.5	2034.0	26
180-199	7	189.5	1326.5	33
200-220	3	210.0	630.0	36
Total	**36**		**5963.5**	

Mean:

$$\bar{x} \approx \frac{\sum fm}{\sum f} = \frac{5963.5}{36} \approx 165.7 \text{ lb}$$

The mean is in the 160-179 class but nearer 160, because more of the data lies on the lower (left) side.

Mode: the highest frequency is 12, so the modal class is **160-179**.

Median: half of 36 is 18. The classes cover students 1-6, 7-14, then **15-26**, so the 18th student is in **160-179**.

Key takeaways

- Grouped data → **estimated** mean: $\sum fm / \sum f$, using class midpoints.
- **Modal class** = highest frequency.
- **Median class** = the class whose cumulative frequency first reaches $n/2$.
- To find the exact median or mode *within* a class, see Lessons 3.16 and 3.17.